

Quang Anh Vu

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EDUCATION

Hanoi University of Science and Technology (HUST)

B.Sc. in Mechatronics Engineering

Hanoi, Vietnam

Aug. 2021 – Aug. 2025

- GPA: 4.0/4.0, CPA: 3.4/4.0.
- Awarded the **Academic Excellence Scholarship** for three terms.

RESEARCH EXPERIENCE

Vinmotion

Robotic Researcher

Hanoi, Vietnam

May 2025 – Present

- Research and develop **deep reinforcement learning (DRL)** algorithms for robot control, focusing on **motion imitation (mimic)** and **locomotion**.
- Deploy learned policies on **physical robot hardware**, systematically closing the **sim-to-sim** and **sim-to-real** gaps.
- Build pipelines that convert human motion data from **video, OptiTrack, and mocap suits** into robot-feasible references via **motion retargeting**.
- Integrate optimization-based controllers on hardware, including **Model Predictive Control (MPC)** and **whole-body control**.
- Develop **teleoperation, perceptive locomotion, and perceptive tracking** systems for real-robot deployment.

CVMR LAB

Undergraduate Research Assistant

Hanoi, Vietnam

September 2023 – May 2025

- Developed **perception and navigation** algorithms enabling mobile robots to plan paths and avoid obstacles autonomously.
- Worked with **robotic arms**, implementing control strategies and deploying them from simulation to real hardware.

PUBLICATIONS

Ukemi-SafeFall: Humanoid Safe Falling & Recovery (*first author*)
Project Page | VinMotion Hardware Demo (0:04–0:22)

Paper Under Review, HUMANOIDS 2026

- Developed a **PPO-based whole-body fall-and-recovery policy** for a **23-DoF, 32 kg humanoid** using **mjlab / MuJoCo**, combining condition-gated impact-mitigation rewards with a **Score-Matching Motion Prior (SMP)**.
- Scaled training to **4,096 parallel environments** and improved robustness using **Generative State Initialization (GSI)**, domain randomization, observation noise, and randomized external force and velocity disturbances.
- In simulation, achieved **98% recovery success** with a **1.04 s mean time-to-stand**; reduced normalized peak contact force by **32.1%** and cumulative peak contact loading by **78.4%** versus passive falling.
- Deployed the learned policy to physical humanoid hardware using a **C++ 50 Hz control stack** with **ONNX Runtime**, demonstrating sim-to-real fall recovery under kicks, obstacle traversal, and elevated-platform falls.

AR-FlowZ: Autoregressive Riemannian Latent Flows for Latent Skill Sequencing in Behavioral Foundation Models (*first author*)
Project Page

Paper Under Review, ICRA 2027

- Proposed **AR-FlowZ**, an **autoregressive Riemannian latent-flow sequencer** on top of a frozen **FB-CPR** behavioral foundation model, generating smooth transitions between latent skills conditioned on state, previous latent, goal, and recurrent history, with a **SLERP-centered geometric prior** on the normalized latent sphere.
- **Distilled** the multi-step flow into a **one-step endpoint predictor**, cutting inference from ~ 100 ms to **3.4 ms** ($\sim 29\times$ faster) to meet the 20 ms budget of a **50 Hz** control loop.
- On a 46 s behavior-switching benchmark (64 envs $\times$ 3 seeds), improved transition success to **78.8%** (vs. 75.9% direct readout) while reducing peak and p99 action rates by **18.8%** and **15.8%**, respectively.
- Deployed on a **Digit humanoid** through a **ten-link Cartesian interface** driven by a mocap suit or monocular camera, with no per-frame inverse kinematics; mean tracking error of **3.4°** (arms) and **11.9°** (legs).

Research on Modeling and Object Tracking for Robot Arm Based on Deep Reinforcement Learning

MM Science Journal, June 2025

- Anh-Dung Nguyen, Tien-Dat Vu, **Quang-Anh Vu**, Thai-Viet Dang | doi.org/10.17973/MMSJ.2025_06_2025059
- Proposed a **PPO**-based robotic-arm control pipeline combined with a **ResNet18-UNet** model for 2D object localization, enabling obstacle-aware manipulation of objects to target positions; validated in the **PyBullet** simulation environment via cumulative-reward optimization.

Optimal Obstacle Avoidance Strategy Using Deep Reinforcement Learning Based on Stereo Camera

MM Science Journal, October 2024

- Chi-Hung Nguyen, **Quang-Anh Vu**, Kim-Khoi Phung Cong, Thai-Viet Dang | doi.org/10.17973/MMSJ.2024_10_2024078
- Applied **YOLOv8**-based stereo-camera obstacle detection with a **Deep Q-Network** to learn collision-free navigation policies for a mobile robot toward a target destination; validated in **ROS2** simulation.

PROJECTS

Full-Size Humanoid Locomotion & Motion Imitation (*1.7m robot*) Hardware Demo (0:10–0:24)

2026

- Developed and deployed **deep RL locomotion** and **whole-body motion-imitation (mimic)** policies on a **1.7 m full-size humanoid**, from simulation training through real-hardware execution.
- Closed the **sim-to-sim** and **sim-to-real** gaps to achieve stable, human-like motion on physical hardware.

Language-Driven Humanoid Control (Text to Robot) ARDY → GMR → BFM-Zero

2026

- Built an end-to-end **text-to-robot pipeline** in which a **single natural-language prompt** is turned into human motion by **ARDY** (NVIDIA's autoregressive diffusion text-to-motion model), eliminating the need for motion-capture recording.
- Retargeted the generated human motion to the robot's kinematics with **General Motion Retargeting (GMR)**, and used the **BFM-Zero behavioral foundation model** as the whole-body controller to imitate the retargeted references.
- **Deployed and ran the full pipeline on the physical Motion Mini humanoid**, letting the robot perform new motions directly from free-form language commands.

TECHNICAL SKILLS

Robot Learning: Deep RL (PPO, DQN), behavioral foundation models (FB-CPR, BFM-Zero), motion imitation & retargeting (GMR), text-to-motion (ARDY), sim-to-real transfer, domain randomization, motion priors

Control & Perception: Model Predictive Control, whole-body control, teleoperation, perceptive locomotion, stereo vision (YOLOv8, ResNet-UNet)

Simulation: MuJoCo, IsaacLab, mjlab, PyBullet, ROS2

Hardware & Sensing: Humanoid robots, robotic arms, OptiTrack, mocap suits

Languages & Deployment: Python, C++, ONNX Runtime, Git